

COR-STAT1202 Introductory Statistics

Tutorial 3

Probability

Q1. Given that $X = \{1, 2, 4, 5\}$ and that $P(X = 1) = 0.2$, $P(X = 2) = 0.5$, $P(X = 4) = 0.2$, $P(X = 5) = 0.1$.

Calculate the following:

a) $P(X^2 \leq 10)$

b) $P(X > 1 \mid X \text{ is less than } 4)$

Q2. Mr Tan, a HDB approved contractor, has submitted two bids for two projects; Project X and Project Y. The probability of getting project X and Project Y is 0.60 and 0.75 respectively. The probability of him getting none of the projects is 0.15.

a) What is the probability that Mr Tan will get both projects?

b) Are the events of getting the two projects mutually exclusive? Why?

c) Are the events of getting the two projects independent? Why?

d) If Mr Tan gets project X, what is the probability that he will get project Y?

e) If Mr Tan gets project Y, what is the probability that he will get project X?

Q3. What is the probability of drawing a king or a spade from a pack of 52 poker cards? cards?

Q4. Tiger Fund Ltd. operates a number of mutual funds in high technology and in financial sectors. Roberts is a fund manager who runs a major fund that includes a wide variety of technology stocks. As fund manager he decides which stocks should be purchased for the mutual fund. The compensation plan for fund managers includes a first-year bonus for each stock purchased by the manager that gains more than 10% in the first six months it is held.

Of those stocks that the company holds, 40% are up in value after being held for two years. In reviewing the performance of Roberts, they found that he received a first-year bonus for 60% of the stocks that he purchased that were up after two years. He also received a first-year bonus for 40% of the stocks he purchased that were not up after two years.

What is the probability that a stock will be up after two years given that Roberts received a first-year bonus?

Q5. A student can enter a course either as a beginner (73%) or as a transferring student (27%). It is found that 62% of beginners eventually graduate, and that 78% of transfers eventually graduate. Find:

a) the probability that a randomly chosen student is a beginner and eventually graduate.

b) the probability that a randomly chosen student will eventually graduate.

c) the probability that a randomly chosen student is either a beginner or will eventually graduate, or both.

d) If a student eventually graduates, what is the probability that the student entered as a transferring student?

e) If two entering students are chosen at random, what is the probability that not only do they enter in the same way but that they also both graduate or fail?

COR-STAT1202 Introductory Statistics

Tutorial 3

Probability

Extra Questions

1. In an experiment, three dice were thrown into five cylinders at random. Assuming each dice has the same chance of going into any cylinder independently of wherever the other 2 dice fall. Determine the probability of 0, 1, 2, 3, 4, and 5 empty cylinders.
2. a) What is the probability of at least one odd number if two fair dice are thrown?
b) What is the probability of at least one even number if three fair dice are thrown?
3. A fair dice is thrown twice. Determine the following:
 - a) A one turns up each time
 - b) The sum of the scores is 9
 - c) Both are odd
4. The number of male and female students taking different modules are shown in the table below.

Module	Male (M)	Female (F)
Statistics (S)	5	14
Economics (E)	17	41
Accounting (A)	4	25
Programming (P)	23	11
Total	49	91

- a) Discuss and justify if the event of taking Statistics and being female are independent events.
 - b) Given that a student takes Economics, what is the probability of this student is a male.
5. The badminton player, Loh Kean Yew, plays two games against an opponent. He has a 50% chance of winning the first game, and if he wins the first, he then has a 55% chance of winning the second game. But if he loses the first, he has only a 40% chance of winning the second game.
 - a) What is the probability of him winning both games?
 - b) If he lost the second game, what is the probability of he lost both games?